BLE_att_client Application Note

Revision History

Date	Version	Description
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Table of Contents

1 Overview 4

2 Quick Start and Flow 4

3 Hierarchy 10

4 Button 10

5 Parameter 11
- 5.1 GAP 11
- 5.2 Scan 11
- 5.3 Init 11
- 5.4 Advertisement 12
- 5.5 Idle Timeout 12

6 Application State 12

7 Client Role 13
- 7.1 Maximum Link as Client Role 13
- 7.2 How to Search BLE_att_server 13
- 7.3 Gatt Client 13

8 Server Role 14

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List of Figures

Figure 1 - Booting Message in Console Log
Figure 2 - Power On Message in Console Log
Figure 3 - Start Scan and Scan Setting Message in Console Log
Figure 4 - Idle Timeout in Console Log
Figure 5 - Connect with BLE_att_server in Console Log
Figure 6 - KEY Action in Console Log (BLE_att_client)
Figure 7 - Received Open Door Command in Console Log (BLE_att_server)
Figure 8 - Enter ADV Mode in Console
Figure 9 - Search from Mobile APP
Figure 10 - Connected with Mobile APP in Console
Figure 11 - Connected with Mobile APP
Figure 12 - Mobile APP Send Write “close door” Command in Console Log
Figure 13 - Hierarchy of BLE_att_client Example
Figure 14 - BLE_att_client Application State

List of Tables

Table 1 - KEY Mapping and Action Table
Table 2 - Default Parameters for Advertisement

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1 Overview

This application note describes the settings, functionality, and code flow of the BLE_att_client example code running on an Atmosic-based Bluetooth LE system. This BLE_att_client will try to search and connect with Atmosic BLE_att_server example devices after power on. When a connection is made, users can trigger with a key to send commands through GATT services. The maximum connection links as client roles are 4. The client device would keep searching until all client links are connected. For applications, it may enter hibernation after idle timeout (no connection state over 30 seconds).

In addition, the example code also can support one link as server role for Mobile APP with GATT services. It can be triggered with a key action to enter advertising mode and then can be connected from the phone. As it entered advertising mode, it would stop searching.

2 Quick Start and Flow

- Hardware setting for Key please refer to section 4 Button.
- BLE_att_client: make run_all USER_BD_ADDR:="a5 a5 a4 33 55 a8"
  "a5 a5 a4 33 55 a8": modify the BD address
- Please prepare another Atmosic evaluation kit (EVK) with BLE_att_server example for testing
- Console Message for different stages
  - After booting

```
@00000033 ATM2201-x0x silicon: 5x5 pkg, External flash
5x5 EXT_FLASH: 4e 56 44 53 05 06 03 50 46 00 06 06 27 00 00 00 ...
@00000237 [BLE_att_cl][D]: user_appm_init:
@000002f5 Button init!!
```

Figure 1 - Booting Message in Console Log

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- Power on (press and hold the KEY1 button over half sec)

```
@004e1dbd key_mask 1,: press 500 ms
@004e1e4b [BLE_att_cl][V]: client_app_key_handler: idx:0, 1
@004e1f21 [BLE_att_cl][D]: client_op_init: con_max:5, att_client:4
@004e200b [ atts_gatt][D]: atts_create_prf:
@004e20db [ atts_gatt][D]: atts_create_prf: svc1 (0), RW1 (2), NT1 (4) NT_CCCD1
(5)
@004e21fa [ atts_gatt][D]: atts_create_prf: svc2 (6), IND2 (8) NT_CCCD2 (9)
@004e22ff [ atts_gatt][D]: atts_create_prf: svc3 (10), RW3 (12), CUDD3 (13)
@004e2416 [ atts_gatt][D]: atts_create_prf: SVC_ECHO(17), RW_ECHO(19), NT_ECHO(2
1), NT_CCCD_ECHO(22)
@004e25be [ble_gap_se][N]: BOND MASK : 0
@004e2718 [  atm_prfs][V]: atm_prfs_init:
@004e27be [   atm_gap][W]: Unhandled GAPM msg 0xd1c
@004e287b [BLE_att_cl][D]: client_gap_init_cfm: status: 0
@004e2949 [BLE_att_cl][D]: client_op_create_scan:
@004e2a43 [ble_atmprf][N]: ble_atmprfs_active_svc_db: svc_idx (0), start_hdl (11
), attr_num (6)
@004e2ba5 [ble_atmprf][N]: ble_atmprfs_active_svc_db: svc_idx (1), start_hdl (17
), attr_num (4)
@004e2d30 [ble_atmprf][N]: ble_atmprfs_active_svc_db: svc_idx (2), start_hdl (21
), attr_num (7)
@004e2ea7 [ble_atmprf][N]: ble_atmprfs_active_svc_db: svc_idx (3), start_hdl (28
), attr_num (6)
@004e2ffb [BLE_att_cl][V]: client_scan_create_cfm: act_idx (0), status (0)
@004e30fe [BLE_att_cl][D]: client_op_start_scan:,scan idx:0
@004e31d0 [  atm_scan][D]: atm_scan_start
@004e326c [  atm_scan][D]:  - period    = 0 ms
@004e3318 [  atm_scan][D]:  - duration  = 0 ms
@004e33c4 [  atm_scan][D]:  - scan_intv = 100 ms
@004e3479 [  atm_scan][D]:  - scan_wd   = 20 ms
@004e3535 [  atm_init][V]: atm_gapm_cmp_evt_create_init(1): status 0
@004e3624 [BLE_att_cl][V]: client_gap_atm_ini_cb: state (0x2) act_idx (1) status
(0)
@004e3783 [BLE_att_cl][V]: client_scan_start_cfm: act_idx (0), status (0)
```

Figure 2 - Power On Message in Console Log

Scan parameter:

```
@004e31d0 [  atm_scan][D]: atm_scan_start
@004e326c [  atm_scan][D]:  - period    = 0 ms
@004e3318 [  atm_scan][D]:  - duration  = 0 ms
@004e33c4 [  atm_scan][D]:  - scan_intv = 100 ms
@004e3479 [  atm_scan][D]:  - scan_wd   = 20 ms
```

Figure 3 - Start Scan and Scan Setting Message in Console Log

- Idle timeout to hibernation mode(Default:30s)

```
@005d2a12 [BLE_att_cl][D]: client_op_idle_timeout
@005d2ae3 [BLE_att_cl][V]: client_scan_stop_ind: act_idx (0), status (0)
@005d2bdf [  atm_scan][D]: Scan0: stopped (0)
```

Figure 4 - Idle Timeout in Console Log

- Connected with BLE_att_server (you can prepare four BLE_att_server)

```
@0001dac1 [BLE_att_cl][D]: client_gap_create_connect:BDADDR:a7,a5,a4,33,55,a8
@0001dbd5 [BLE_att_cl][V]: client_gap_ext_adv_ind:head:1, 1, bd:0xa7
@0001dcee [  atm_init][D]: Page(1): ON
@0001dd87 [BLE_att_cl][V]: client_gap_atm_ini_cb: state (0x5) act_idx (1) status(0)
@0001fa0d [  atm_prfs][V]: atm_prfs_create: conidx (0), task (10), id (118)
@0001fb21 [BLE_att_cl][D]: client_gap_conn_ind: conidx-0, idx-0
@0001fc05 [BLE_att_cl][D]: client_op_connected current:0, total:1
@0001fcea Conidx:0: att_client_disc_handle
@0001fd89 [ attc_gatt][V]: att_client_disc_handle: conidx (0), attr_idx (0)
@0001fe92 [   atm_gap][D]: + Peer (0) A8:55:33:A4:A5:A7
@0001ff62 [   atm_gap][D]: + Connection interval +    16 (unit:1.25ms)
@00020056 [   atm_gap][D]: + Slave latency       +    10
@00020120 [   atm_gap][D]: + Supervision timeout +   500 (unit:  10ms)
@0002021a [   atm_gap][D]: ConnInd idx: 0 role: M
@000202e1 [BLE_att_cl][V]: client_gap_atm_ini_cb: state (0x3) act_idx (1) status(0)
@00020401 [BLE_att_cl][D]: client_op_wait_next
@000204ab [  atm_init][D]: Page(1): OFF (reason: 0)
@00020584 [   atm_gap][W]: Unhandled GAPC msg 0xe2c
@00021e83 Conidx:0: att_client_disc_svc_cb
@00021f21 [ attc_gatt][V]: att_client_disc_svc_cb:0 shdl (11), ehdl (16)
@00023e79 [BLE_att_cl][D]: client_op_connecting:,tail:2, 1
@00023f50 [BLE_att_cl][D]: client_gap_create_connect:BDADDR:a6,a3,a4,33,66,a9
@00024064 [BLE_att_cl][V]: client_gap_ext_adv_ind:head:2, 2, bd:0xa6
@00024180 [  atm_init][D]: Page(1): ON
@00024215 [BLE_att_cl][V]: client_gap_atm_ini_cb: state (0x5) act_idx (1) status(0)
@00027f3f [  atm_prfs][V]: atm_prfs_create: conidx (1), task (10), id (118)
@00028053 [BLE_att_cl][D]: client_gap_conn_ind: conidx-0x1, idx-0x1
@00028142 [BLE_att_cl][D]: client_op_connected current:1, total:2
@00028227 Conidx:1: att_client_disc_handle
@000282c6 [ attc_gatt][V]: att_client_disc_handle: conidx (1), attr_idx (0)
@000283cf [   atm_gap][D]: + Peer (0) A9:66:33:A4:A3:A6
@000284a0 [   atm_gap][D]: + Connection interval +    16 (unit:1.25ms)
@00028596 [   atm_gap][D]: + Slave latency       +    10
@00028660 [   atm_gap][D]: + Supervision timeout +   500 (unit:  10ms)
@00028757 [   atm_gap][D]: ConnInd idx: 1 role: M
@00028817 [BLE_att_cl][V]: client_gap_atm_ini_cb: state (0x3) act_idx (1) status(0)
@00028937 [BLE_att_cl][D]: client_op_wait_next
@000289e1 [  atm_init][D]: Page(1): OFF (reason: 0)
@00028ac9 [   atm_gap][W]: Unhandled GAPC msg 0xe2c
@00028f1a [BLE_att_cl][D]: client_op_connecting:,tail:3, 2
@00028fec [BLE_att_cl][D]: client_gap_create_connect:BDADDR:a2,a2,a4,44,88,a2
@00029101 [BLE_att_cl][V]: client_gap_ext_adv_ind:head:3, 3, bd:0xa2
@00029219 [  atm_init][D]: Page(1): ON
@000292b2 [BLE_att_cl][V]: client_gap_atm_ini_cb: state (0x5) act_idx (1) status(0)
@00029990 Conidx:1: att_client_disc_svc_cb
@00029a2e [ attc_gatt][V]: att_client_disc_svc_cb:1 shdl (11), ehdl (16)
@0002b070 [  atm_prfs][V]: atm_prfs_create: conidx (2), task (10), id (118)
@0002b184 [BLE_att_cl][D]: client_gap_conn_ind: conidx-0x2, idx-0x2
@0002b273 [BLE_att_cl][D]: client_op_connected current:2, total:3
@0002b359 Conidx:2: att_client_disc_handle
@0002b3f7 [ attc_gatt][V]: att_client_disc_handle: conidx (2), attr_idx (0)
@0002b500 [   atm_gap][D]: + Peer (0) A2:88:44:A4:A2:A2
@0002b5d1 [   atm_gap][D]: + Connection interval +    16 (unit:1.25ms)
@0002b6c5 [   atm_gap][D]: + Slave latency       +    10
@0002b78f [   atm_gap][D]: + Supervision timeout +   500 (unit:  10ms)
@0002b886 [   atm_gap][D]: ConnInd idx: 2 role: M
@0002b940 [BLE_att_cl][V]: client_gap_atm_ini_cb: state (0x3) act_idx (1) status(0)
@0002ba60 [BLE_att_cl][D]: client_op_wait_next
@0002bb0a [  atm_init][D]: Page(1): OFF (reason: 0)
@0002bbea [   atm_gap][W]: Unhandled GAPC msg 0xe2c
@0002c542 Conidx:2: att_client_disc_svc_cb
@0002c5e0 [ attc_gatt][V]: att_client_disc_svc_cb:2 shdl (11), ehdl (16)
@0002f304 Conidx:0: att_client_disc_descs_cb
@0002f3a7 [ attc_gatt][E]: att_client_disc_descs_cb: status (a)
@0002f486 [ attc_gatt][V]: att_client_disc_descs_cb: conidx:0 att_idx (1) hdl (13)
@0002f5a9 [ attc_gatt][V]: att_client_disc_descs_cb: conidx:0 att_idx (2) hdl (15)
@0002f6c2 [ attc_gatt][V]: att_client_disc_descs_cb: conidx:0 att_idx (3) hdl (16)
@0002f7dd Conidx:0: att_client_disc_handle
@0002f87b [ attc_gatt][V]: att_client_disc_handle: conidx (0), attr_idx (4)
@000318a7 [BLE_att_cl][D]: client_op_connecting:,tail:4, 3
@00031979 [BLE_att_cl][D]: client_gap_create_connect:BDADDR:a5,a5,a4,33,55,a8
@00031a93 [BLE_att_cl][V]: client_gap_ext_adv_ind:head:4, 4, bd:0xa5
@00031bbc [  atm_init][D]: Page(1): ON
@00031c50 [BLE_att_cl][V]: client_gap_atm_ini_cb: state (0x5) act_idx (1) status(0)
@00031d80 Conidx:0: att_client_disc_svc_cb
@00031e1e [ attc_gatt][V]: att_client_disc_svc_cb:0 shdl (17), ehdl (20)
@0003784d Conidx:1: att_client_disc_descs_cb
@000378f1 [ attc_gatt][E]: att_client_disc_descs_cb: status (a)
@000379d0 [ attc_gatt][V]: att_client_disc_descs_cb: conidx:1 att_idx (1) hdl (13)
@00037aeb [ attc_gatt][V]: att_client_disc_descs_cb: conidx:1 att_idx (2) hdl (15)
@00037c04 [ attc_gatt][V]: att_client_disc_descs_cb: conidx:1 att_idx (3) hdl (16)
@00037d1f Conidx:1: att_client_disc_handle
@00037dbd [ attc_gatt][V]: att_client_disc_handle: conidx (1), attr_idx (4)
@00039bda [  atm_prfs][V]: atm_prfs_create: conidx (3), task (10), id (118)
@00039cee [BLE_att_cl][D]: client_gap_conn_ind: conidx-0x3, idx-0x3
@00039ddd [BLE_att_cl][D]: client_op_connected current:3, total:4
@00039ec3 Conidx:3: att_client_disc_handle
@00039f62 [ attc_gatt][V]: att_client_disc_handle: conidx (3), attr_idx (0)
@0003a06b [BLE_att_cl][D]: client_op_connected_all
@0003a125 [   atm_gap][D]: + Peer (0) A8:55:33:A4:A5:A5
@0003a1f6 [   atm_gap][D]: + Connection interval +    16 (unit:1.25ms)
@0003a2ea [   atm_gap][D]: + Slave latency       +    10
@0003a3b8 [   atm_gap][D]: + Supervision timeout +   500 (unit:  10ms)
```

Figure 5 - Connect with BLE_att_server in Console Log

- Send write “open door” command to BLE_att_server (tap KEY1 ~ 4, KEY1 for link0 and KEY2 for link1 etc.)

```
@00db3a41 key 0,: tap 124 ms
@00db3abd [BLE_att_cl][V]: client_app_key_handler: idx:0, 0
@00db3b92 [ attc_gatt][V]: att_client_enable_ntf_ind:conidx(0)-15 cccd(0x1)
@00db5729 [ attc_gatt][D]: att_client_sent_write_cb: status:0
@00db8ce1 [ attc_gatt][D]: att_client_sent_write_cb: status:0
@00db8dd4 [ attc_gatt][V]: attc_gattc_event_ind: conidx (0), attr:14
@00dbd92a key 1,: tap 125 ms
@00dbd9a4 [BLE_att_cl][V]: client_app_key_handler: idx:1, 0
@00dbda79 [ attc_gatt][V]: att_client_enable_ntf_ind:conidx(1)-15 cccd(0x1)
@00dbfdb9 [ attc_gatt][D]: att_client_sent_write_cb: status:0
@00dc3371 [ attc_gatt][D]: att_client_sent_write_cb: status:0
@00dc3464 [ attc_gatt][V]: attc_gattc_event_ind: conidx (1), attr:14
@00dd308b key 2,: tap 90 ms
@00dd3100 [BLE_att_cl][V]: client_app_key_handler: idx:2, 0
@00dd31d5 [ attc_gatt][V]: att_client_enable_ntf_ind:conidx(2)-15 cccd(0x1)
@00dd4301 [ attc_gatt][D]: att_client_sent_write_cb: status:0
@00dd78b9 [ attc_gatt][D]: att_client_sent_write_cb: status:0
@00dd79ac [ attc_gatt][V]: attc_gattc_event_ind: conidx (2), attr:14
@00ddcd93 key 3,: tap 107 ms
@00ddce0c [BLE_att_cl][V]: client_app_key_handler: idx:3, 0
@00ddcee1 [ attc_gatt][V]: att_client_enable_ntf_ind:conidx(3)-15 cccd(0x1)
@00dde991 [ attc_gatt][D]: att_client_sent_write_cb: status:0
@00de1a2a [ attc_gatt][D]: att_client_sent_write_cb: status:0
@00de1b1d [ attc_gatt][V]: attc_gattc_event_ind: conidx (3), attr:14
```

Figure 6 - KEY Action in Console Log (BLE_att_client)

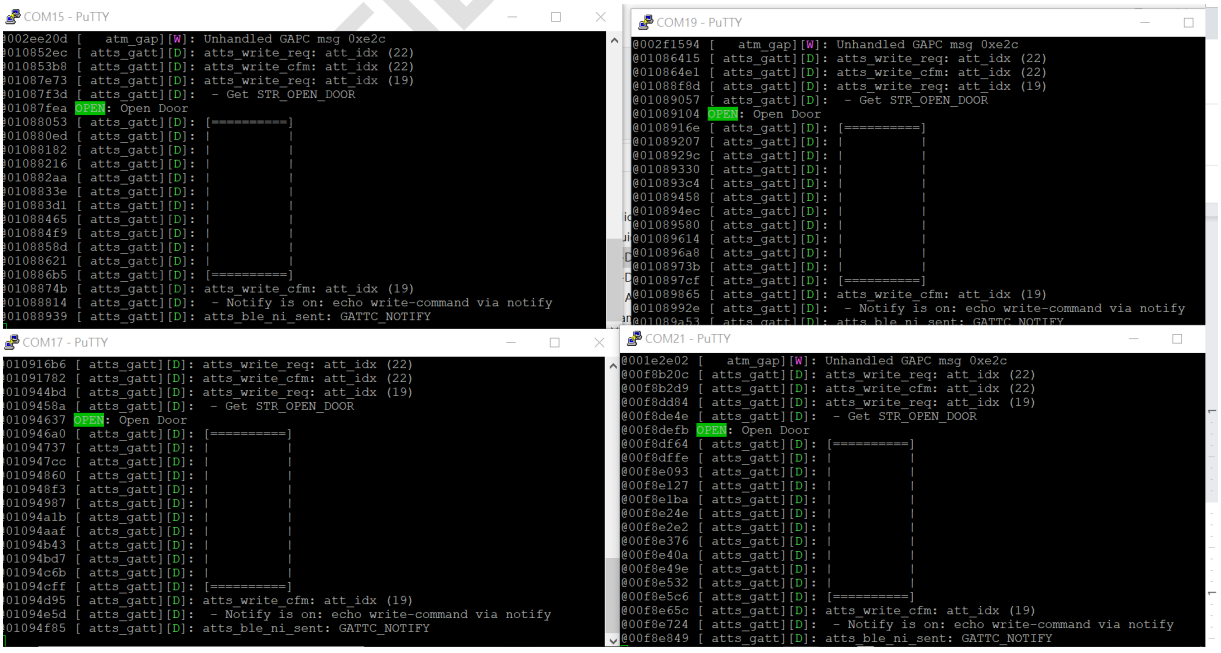


Figure 7 - Received Open Door Command in Console Log (BLE_att_server)

- Enter ADV mode (press and hold the KEY2 button over half sec)

```
@0173ad09 key_mask 2,: press 500 ms
@0173ad98 [BLE_att_cl][V]: client_app_key_handler: idx:1, 1
@0173ae6f [BLE_att_cl][D]: client_op_create_adv:
@0173af27 [BLE_att_cl][D]: client_op_create_adv: adv_param.prop (3)
@0173b033 [BLE_att_cl][D]: ATM_ADV_CREATED(0)
@0173b0dc [BLE_att_cl][V]: client_adv_create_cfm: idx = 2
@0173b1c8 [BLE_att_cl][D]: ATM_ADV_ADVDATA_DONE(0)
@0173b297 [BLE_att_cl][D]: ATM_ADV_SCANDATA_DONE(0)
@0173b375 [   atm_adv][D]: Adv2: ON (0)
@0173b40e [BLE_att_cl][D]: ATM_ADV_ON
```

Figure 8 - Enter ADV Mode in Console

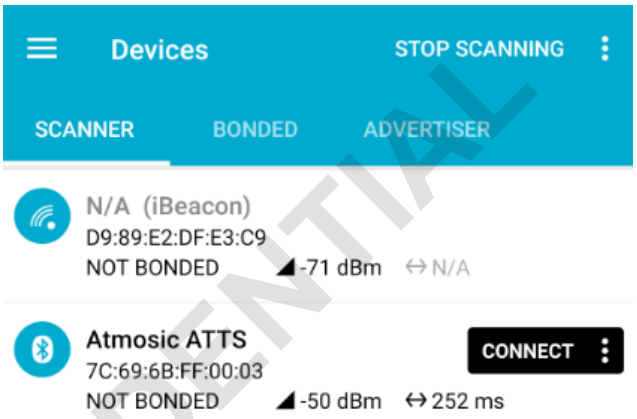


Figure 9 - Search from Mobile APP

- Connect with Mobile APP

```
@01acba70 [  atm_prfs][V]: atm_prfs_create: conidx (4), task (10), id (118)
@01acbb89 [   atm_gap][D]: + Peer (1) 78:76:BB:A4:D2:53
@01acbc5d [   atm_gap][D]: + Connection interval +    39 (unit:1.25ms)
@01acbd50 [   atm_gap][D]: + Slave latency       +     0
@01acbe19 [   atm_gap][D]: + Supervision timeout +   500 (unit:  10ms)
@01acbf19 [   atm_gap][D]: ConnInd idx: 4 role: S
@01acbfd3 [   atm_adv][D]: Adv2: OFF (0)
@01acc070 [BLE_att_cl][D]: ATM_ADV_OFF
@01acc108 [BLE_att_cl][D]: client_op_adv_off: total:4
@01acc1e6 [BLE_att_cl][D]: ATM_ADV_DELETED
```

Figure 10 - Connected with Mobile APP in Console

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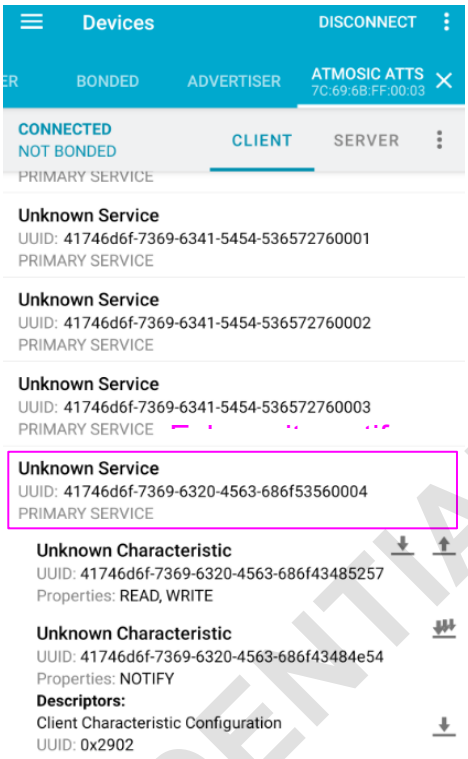


Figure 11 - Connected with Mobile APP

- Mobile APP send write “close door” command

```
@01acc1e6 [BLE_att_cl][D]: ATM_ADV_DELETED
@02d49f32 [ atts_gatt][D]: atts_write_req: att_idx (22)
@02d49ffe [ atts_gatt][D]: atts_write_cfm: att_idx (22)
@02db5f9c [ atts_gatt][D]: atts_write_req: att_idx (19)
@02db6066 [ atts_gatt][D]:  - Get STR_CLOSE_DOOR
@02db6116 CLOSE: Close Door
@02db6185 [ atts_gatt][D]: [==========]
@02db621b [ atts_gatt][D]: |x        x|
@02db62b0 [ atts_gatt][D]: | x      x |
@02db6345 [ atts_gatt][D]: |  x    x  |
@02db63da [ atts_gatt][D]: |   x  x   |
@02db646f [ atts_gatt][D]: |    xx    |
@02db6504 [ atts_gatt][D]: |    xx    |
@02db6598 [ atts_gatt][D]: |   x  x   |
@02db662e [ atts_gatt][D]: |  x    x  |
@02db66c2 [ atts_gatt][D]: | x      x |
@02db6756 [ atts_gatt][D]: |x        x|
@02db67eb [ atts_gatt][D]: [==========]
@02db6880 [ atts_gatt][D]: atts_write_cfm: att_idx (19)
@02db6949 [ atts_gatt][D]:  - Notify is on: echo write-command via notify
@02db6a74 [ atts_gatt][D]: atts_ble_ni_sent: GATTC_NOTIFY
```

Figure 12 - Mobile APP Send Write “close door” Command in Console Log

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3 Hierarchy

For BLE_att_client examples, below is the hierarchy of related modules. The top level is BLE_att_client and it interacts with all other modules according to user scenarios.

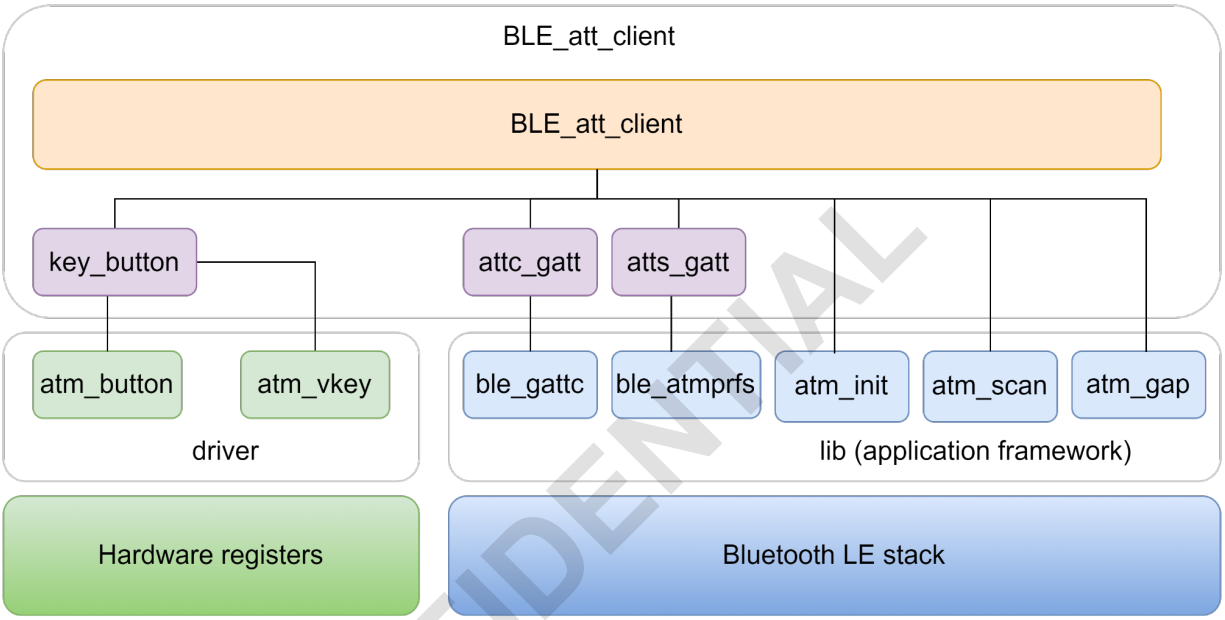


Figure 13 - Hierarchy of BLE_att_client Example

4 Button

There are four keys and 6 key actions. For related PIO definition and action handler, it is implemented in key_button.c/key_button.h.

KEY	KEY1	KEY2	KEY3	KEY4
PIN	P9	P10	P25	P30
Action	Tap: gatt command to link0 LongPress(2s): Power on	Tap: gatt command to link1 Press: ADV mode on	Tap: gatt command to link2	Tap: gatt command to link3

Table 1 - KEY Mapping and Action Table

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5 Parameter

5.1 GAP

The default parameter is defined in “param_gap.hattc_param_gap.h”
For the application, it can be as server and client, hence set the role to GAP_ROLE_ALL.

```
#define CFG_GAP_ROLE GAP_ROLE_ALL
```

5.2 Scan

The default parameter is defined in “attc_param_gap.h”
For the application, GAPM_SCAN_PROP_ACTIVE_1M_BIT is required. Please do not disable it.

```
#define CFG_SCAN0_START_PROP
      (GAPM_SCAN_PROP_PHY_1M_BIT | GAPM_SCAN_PROP_ACTIVE_1M_BIT)

/// SCAN0 scan interval(Max:10.24s)
#define SCAN0_INTERVAL_CS 10
/// SCAN0 scan window(Max:10.24s)
#define SCAN0_WINDOW_CS 2
/// SCAN0 parameter, unit:625us
#define CFG_SCAN0_START_1M_PARAM (SCAN0_INTERVAL_CS<<4), (SCAN0_WINDOW_
```

5.3 Init

The default parameter is defined in “attc_param_gap.h”

```
/// Connection interval for client role
#define CFG_INIT_CONN_PARAM_1M_INTV_MIN 16
#define CFG_INIT_CONN_PARAM_1M_INTV_MAX 16
```

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5.4 Advertisement

The default parameters for advertisement are loaded from Flash Non-Volatile Data Storage (NVDS).

	Device Name	Advertisement Interval	Advertisement Timeout
Default Value	Atmosic ATTS	250 ms	180 secs
Flash NVDS Configuration	tag_data\0c-APP_BLE_SCAN_RESP_DATA	tag_data\06-APP_BLE_ACT_CRT_CMD	tag_data\05-APP_BLE_ACT_STRT_CMD

Table 2 - Default Parameters for Advertisement

5.5 Idle Timeout

The default parameter is defined in “BLE_att_client.c” and the value is 30s. If there is no link after power on for 30s, it would enter hibernation mode.

```
#define CLIENT_IDLE_TIMEOUT_CS 3000
```

6 Application State

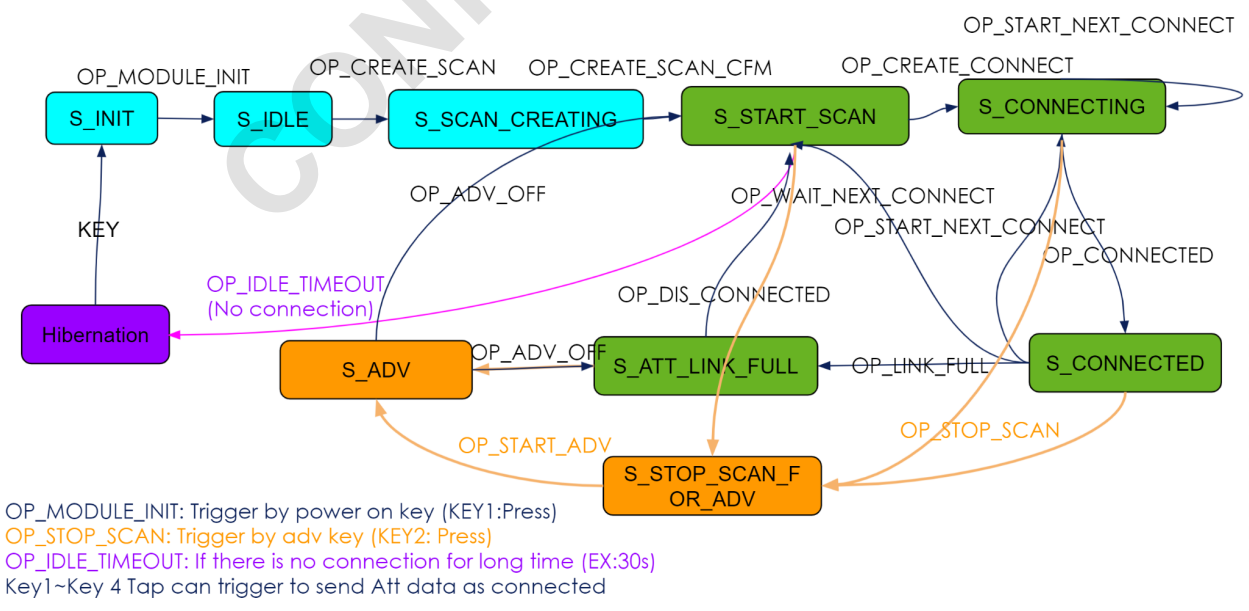


Figure 14 - BLE_att_client Application State

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7 Client Role

7.1 Maximum Link as Client Role

ATT_CLIENT_CONN_MAX is the maximum connect link as client role. The default parameter is defined in “BLE_att_client.c” and it is 4 by default. BLE_att_client devices will try to search BLE_att_server devices until the connected link number is equal to ATT_CLIENT_CONN_MAX. ATT_CLIENT_CONN_MAX needs to be smaller or equal to ATT_CLIENT_LINK_MAX.

7.2 How to Search BLE_att_server

Check the received adv data with specified service UUID as below:

```
/// ADV TYPE for ATT SERVER in ADV DATA
#define ADV_ATT_SERVER_TYPE 0x03
/// UUID for ATT SERVER in ADV DATA
#define ADV_ATT_SERVER_UUID_LOW 0x15
/// UUID for ATT SERVER in ADV DATA
#define ADV_ATT_SERVER_UUID_HIGH 0x18
```

7.3 Gatt Client

ATT_CLIENT_LINK_MAX is the maximum client link that can be handled by attc_gatt. The default value is defined in “attc_gatt.h” and is set to 4. For gatt client, The related APIs are in attc_gatt.c/attc_gatt.h. Client would discover related services supported by BLE_att_server. The service UUID are as below:

```
static uint8_t const atms_uuid[ATT_UUID_128_LEN] = {'A', 't', 'm', 'o', 's',
      'i', 'c', 'A', 'T', 'T', 'S', 'e', 'r', 'v', 0x00, 0x01};

static uint8_t const atms_uuid2[ATT_UUID_128_LEN] = {'A', 't',          'm', 'o', 's',
      'i', 'c', 'A', 'T', 'T', 'S', 'e', 'r', 'v', 0x00, 0x02};

static uint8_t const atms_uuid3[ATT_UUID_128_LEN] = {'A', 't', 'm', 'o', 's',
      'i', 'c', 'A', 'T', 'T', 'S', 'e', 'r', 'v', 0x00, 0x03};

static uint8_t const echo_sv_uuid[AT     T_UUID_128_LEN] = {'A', 't', 'm', 'o', 's',
      'i', 'c', ' ', 'E', 'c', 'h', 'o', 'S', 'V', 0x00, 0x04};
```

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8 Server Role

Users can trigger the mode with KEY2 press. The device will enter advertising mode and then connect from the phone and stop scanning. For the server role, the supported att servers are listed in atts_gatt.c/atts_gatt.h and same with the att servers of BLE_att_server example. For more detail, please refer to the BLE_att_server Application Note (available on the Atomsic Support Website). If ATT_SERVER_CONN_MAX is set to 0, the mode would be disabled. The maximum value for ATT_SERVER_CONN_MAX is 1.

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